

Antiderivatives and Riemann Sums



Basic antiderivatives

1 Find the general antiderivative of each function:

a $f(x) = x^4$

b $f(x) = \frac{1}{x}$

c $f(x) = e^x$

d $f(x) = \sec^2 x$

2 Find $\int (3x^2 - 4x + 5) dx$.

3 Find the particular antiderivative $F(x)$ of $f(x) = 6x^2 - 4$ satisfying $F(1) = 5$.

Riemann sums: left, right, and midpoint

4 Approximate $\int_0^4 x^2 dx$ using a Left Riemann Sum with $n = 4$ equal subintervals.

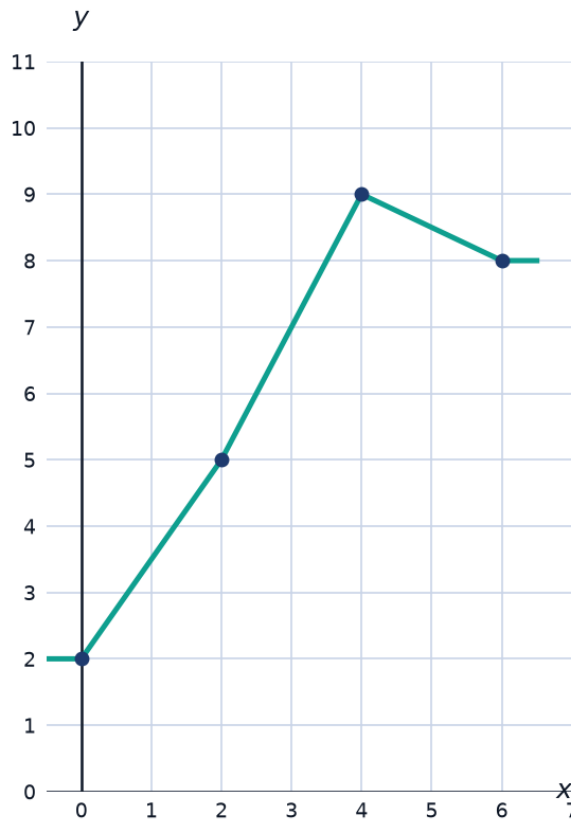
5 Approximate $\int_0^4 x^2 dx$ using a Right Riemann Sum with $n = 4$ equal subintervals.

6 Approximate $\int_0^4 x^2 dx$ using a Midpoint Riemann Sum with $n = 4$ equal subintervals.

The Trapezoidal Rule

7 Use the Trapezoidal Rule with $n = 4$ to approximate $\int_0^4 x^2 dx$.

- 8 A table gives $f(0) = 2$, $f(2) = 5$, $f(4) = 9$, $f(6) = 8$. Use the Trapezoidal Rule with these unequal-width data (widths of 2 each) to approximate $\int_0^6 f(x) dx$.



Sums as limits: the definite integral

- 9 Explain how the definite integral $\int_a^b f(x) dx$ is defined as a limit of Riemann sums, and write the limit expression using sigma notation.
- 10 Determine whether the Left Riemann Sum overestimates or underestimates $\int_0^4 x^2 dx$, and explain why using the shape of the graph.

Riemann sums from a table of values

- 11 A table gives $f(0) = 3$, $f(1) = 5$, $f(2) = 8$, $f(3) = 10$, $f(4) = 9$. Use a Right Riemann Sum with 4 subintervals of width 1 to approximate $\int_0^4 f(x) dx$.

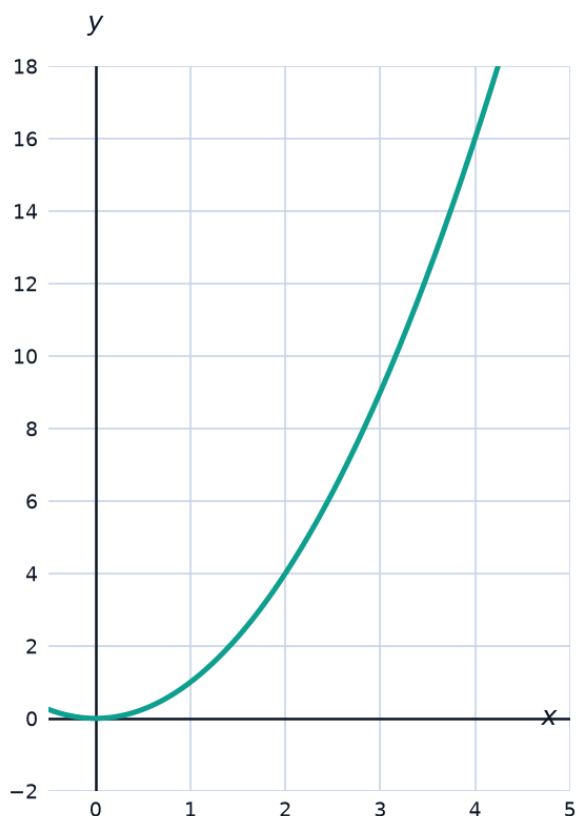
Antiderivatives of trigonometric and exponential functions

- 12 Find $\int (3\sin x + 2e^x) dx$.

- 13 Find the particular antiderivative $F(x)$ of $f(x) = \sec^2 x$ satisfying $F(0) = 2$.
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Visualizing a Riemann sum approximation

- 14 The graph shows $f(x) = x^2$ on $[0, 4]$. Use the graph to explain why the Right Riemann Sum $R_4 = 30$ (found earlier) overestimates the true area, and why the Left Riemann Sum $L_4 = 14$ underestimates it.



An applied antiderivative problem

- 15 A factory's production rate is $r(t) = 100 + 20t$ units/day, where t is days since a process change. Find the total production function $P(t)$ given $P(0) = 0$, then find the total production over the first 10 days.
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More antiderivative practice

- 16 Find the general antiderivative of $f(x) = 4x^3 - \frac{2}{x^2} + \cos x$.
- 17 Find the particular antiderivative $F(x)$ of $f(x) = x^2 - 3$ satisfying $F(0) = 4$.

Comparing Riemann sums to the exact integral

- 18 For $\int_0^4 x^2 dx$, the exact value is $\frac{64}{3} \approx 21.33$. Rank the four approximations found earlier — $L_4 = 14$, $R_4 = 30$, $M_4 = 21$, $T_4 = 22$ — from closest to farthest from the exact value, and explain which type of approximation (midpoint or trapezoidal) tends to be more accurate for a smooth curving function.
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Multiple choice — easy 15

- 19 Find C in the particular antiderivative $F(x) = x^3 - 7x + C$ of $f(x) = 3x^2 - 7$, satisfying $F(-1) = -8$.
- a. -8
 - b. -12
 - c. -14
 - d. -16
- 20 Find C in the particular antiderivative $F(x) = x^3 + 3x + C$ of $f(x) = 3x^2 + 3$, satisfying $F(3) = -5$.
- a. -39
 - b. -43
 - c. -41
 - d. -5
- 21 Find C in the particular antiderivative $F(x) = -2x^3 - 5x + C$ of $f(x) = -6x^2 - 5$, satisfying $F(2) = -9$.
- a. 17
 - b. -9
 - c. 15
 - d. 19

- 22 Find C in the particular antiderivative $F(x) = x^3 + 7x + C$ of $f(x) = 3x^2 + 7$, satisfying $F(-3) = -2$.
- a. 46
 - b. 48
 - c. 44
 - d. -2
- 23 Find C in the particular antiderivative $F(x) = 2x^3 + 9x + C$ of $f(x) = 6x^2 + 9$, satisfying $F(0) = 9$.
- a. 7
 - b. 9
 - c. 10
 - d. 11
- 24 Find C in the particular antiderivative $F(x) = 2x^3 + 3x + C$ of $f(x) = 6x^2 + 3$, satisfying $F(-3) = 6$.
- a. 6
 - b. 69
 - c. 71
 - d. 67
- 25 Find C in the particular antiderivative $F(x) = 2x^3 + x + C$ of $f(x) = 6x^2 + 1$, satisfying $F(3) = -8$.
- a. -65
 - b. -8
 - c. -67
 - d. -63

- 26 Find C in the particular antiderivative $F(x) = -x^3 - 4x + C$ of $f(x) = -3x^2 - 4$, satisfying $F(-3) = 8$.
- a. 8
 - b. -29
 - c. -33
 - d. -31
- 27 Find C in the particular antiderivative $F(x) = x^3 + 5x + C$ of $f(x) = 3x^2 + 5$, satisfying $F(3) = 4$.
- a. 4
 - b. -36
 - c. -38
 - d. -40
- 28 Find C in the particular antiderivative $F(x) = -2x^3 + 7x + C$ of $f(x) = -6x^2 + 7$, satisfying $F(3) = -8$.
- a. 27
 - b. 25
 - c. -8
 - d. 23
- 29 Find C in the particular antiderivative $F(x) = x^3 + x + C$ of $f(x) = 3x^2 + 1$, satisfying $F(-1) = -7$.
- a. -7
 - b. -3
 - c. -4
 - d. -5

- 30 Find C in the particular antiderivative $F(x) = -2x^3 + 6x + C$ of $f(x) = -6x^2 + 6$, satisfying $F(-3) = -8$.
- a. -8
 - b. -46
 - c. -44
 - d. -42
- 31 Find C in the particular antiderivative $F(x) = 2x^3 + 8x + C$ of $f(x) = 6x^2 + 8$, satisfying $F(0) = 5$.
- a. 7
 - b. 5
 - c. 3
 - d. 6
- 32 Find C in the particular antiderivative $F(x) = -x^3 + C$ of $f(x) = -3x^2$, satisfying $F(2) = -1$.
- a. 9
 - b. 5
 - c. -1
 - d. 7
- 33 Find C in the particular antiderivative $F(x) = 2x^3 + 4x + C$ of $f(x) = 6x^2 + 4$, satisfying $F(1) = 2$.
- a. -2
 - b. -6
 - c. 2
 - d. -4

Multiple choice — medium 15

- 34 Approximate $\int_0^4 5x^2 dx$ using a Right Riemann Sum with $n = 4$ equal subintervals.
- a. 150
 - b. 145
 - c. 170
 - d. 70
- 35 Approximate $\int_0^4 6x^2 dx$ using a Right Riemann Sum with $n = 4$ equal subintervals.
- a. 174
 - b. 84
 - c. 180
 - d. 204
- 36 Approximate $\int_0^6 2x^2 dx$ using a Right Riemann Sum with $n = 6$ equal subintervals.
- a. 194
 - b. 110
 - c. 180
 - d. 182
- 37 Approximate $\int_0^7 6x^2 dx$ using a Right Riemann Sum with $n = 7$ equal subintervals.
- a. 840
 - b. 546
 - c. 882
 - d. 834

- 38 Approximate $\int_0^3 5x^2 dx$ using a Right Riemann Sum with $n = 3$ equal subintervals.
- a. 70
 - b. 85
 - c. 65
 - d. 25
- 39 Approximate $\int_0^3 2x^2 dx$ using a Right Riemann Sum with $n = 3$ equal subintervals.
- a. 28
 - b. 34
 - c. 10
 - d. 26
- 40 Approximate $\int_0^4 4x^2 dx$ using a Right Riemann Sum with $n = 4$ equal subintervals.
- a. 120
 - b. 116
 - c. 136
 - d. 56
- 41 Approximate $\int_0^4 3x^2 dx$ using a Right Riemann Sum with $n = 4$ equal subintervals.
- a. 42
 - b. 90
 - c. 102
 - d. 87

- 42 Approximate $\int_0^7 x^2 dx$ using a Right Riemann Sum with $n = 7$ equal subintervals.
- a. 91
 - b. 140
 - c. 147
 - d. 139
- 43 Approximate $\int_0^5 5x^2 dx$ using a Right Riemann Sum with $n = 5$ equal subintervals.
- a. 270
 - b. 275
 - c. 150
 - d. 300
- 44 Approximate $\int_0^7 5x^2 dx$ using a Right Riemann Sum with $n = 7$ equal subintervals.
- a. 455
 - b. 700
 - c. 735
 - d. 695
- 45 Approximate $\int_0^6 x^2 dx$ using a Right Riemann Sum with $n = 6$ equal subintervals.
- a. 55
 - b. 91
 - c. 90
 - d. 97

- 46 Approximate $\int_0^5 3x^2 dx$ using a Right Riemann Sum with $n = 5$ equal subintervals.
- a. 165
 - b. 162
 - c. 90
 - d. 180
- 47 Approximate $\int_0^5 2x^2 dx$ using a Right Riemann Sum with $n = 5$ equal subintervals.
- a. 110
 - b. 120
 - c. 60
 - d. 108
- 48 Approximate $\int_0^4 2x^2 dx$ using a Right Riemann Sum with $n = 4$ equal subintervals.
- a. 68
 - b. 58
 - c. 60
 - d. 28

Multiple choice — hard 15

- 49 Approximate $\int_0^6 f(x) dx$ using the Trapezoidal Rule (equal width 2), given $f(0) = 7$, $f(2) = 12$, $f(4) = 9$, $f(6) = 3$.
- a. 54
 - b. 52
 - c. 62
 - d. 50

- 50 Approximate $\int_0^3 f(x) dx$ using the Trapezoidal Rule (equal width 1), given $f(0) = 7$, $f(1) = 15$, $f(2) = 3$, $f(3) = 9$.
- a. 25
 - b. 27
 - c. 26
 - d. 34
- 51 Approximate $\int_0^3 f(x) dx$ using the Trapezoidal Rule (equal width 1), given $f(0) = 4$, $f(1) = 7$, $f(2) = 12$, $f(3) = 2$.
- a. 22
 - b. 25
 - c. 23
 - d. 21
- 52 Approximate $\int_0^6 f(x) dx$ using the Trapezoidal Rule (equal width 2), given $f(0) = 6$, $f(2) = 4$, $f(4) = 13$, $f(6) = 6$.
- a. 44
 - b. 46
 - c. 48
 - d. 58
- 53 Approximate $\int_0^3 f(x) dx$ using the Trapezoidal Rule (equal width 1), given $f(0) = 1$, $f(1) = 5$, $f(2) = 11$, $f(3) = 9$.
- a. 20
 - b. 21
 - c. 22
 - d. 26

- 54 For $f(x) = 4x^2$ on $[0, 5]$, determine whether the Left and Right Riemann Sums overestimate or underestimate $\int_0^5 f(x) dx$.
- a. Both Left and Right Riemann Sums underestimate.
 - b. Both Left and Right Riemann Sums overestimate.
 - c. Left Riemann Sum overestimates, and Right Riemann Sum underestimates.
 - d. Left Riemann Sum underestimates, and Right Riemann Sum overestimates.
- 55 For $f(x) = -x^2 + 36$ on $[0, 6]$, determine whether the Left and Right Riemann Sums overestimate or underestimate $\int_0^6 f(x) dx$.
- a. Left Riemann Sum overestimates, and Right Riemann Sum underestimates.
 - b. Left Riemann Sum underestimates, and Right Riemann Sum overestimates.
 - c. Both Left and Right Riemann Sums overestimate.
 - d. Both Left and Right Riemann Sums underestimate.
- 56 For $f(x) = 3x^2$ on $[0, 3]$, determine whether the Left and Right Riemann Sums overestimate or underestimate $\int_0^3 f(x) dx$.
- a. Left Riemann Sum overestimates, and Right Riemann Sum underestimates.
 - b. Both Left and Right Riemann Sums overestimate.
 - c. Left Riemann Sum underestimates, and Right Riemann Sum overestimates.
 - d. Both Left and Right Riemann Sums underestimate.
- 57 For $f(x) = 2x^2$ on $[0, 4]$, determine whether the Left and Right Riemann Sums overestimate or underestimate $\int_0^4 f(x) dx$.
- a. Left Riemann Sum overestimates, and Right Riemann Sum underestimates.
 - b. Both Left and Right Riemann Sums underestimate.
 - c. Left Riemann Sum underestimates, and Right Riemann Sum overestimates.
 - d. Both Left and Right Riemann Sums overestimate.

- 58 For $f(x) = -2x^2 + 8$ on $[0, 2]$, determine whether the Left and Right Riemann Sums overestimate or underestimate $\int_0^2 f(x) dx$.
- a. Both Left and Right Riemann Sums underestimate.
 - b. Both Left and Right Riemann Sums overestimate.
 - c. Left Riemann Sum overestimates, and Right Riemann Sum underestimates.
 - d. Left Riemann Sum underestimates, and Right Riemann Sum overestimates.
- 59 A factory's production rate is $r(t) = 80 + 26t$ units/day, where t is days since a process change. Find the total production $P(2)$ given $P(0) = 0$.
- a. 160
 - b. 238
 - c. 186
 - d. 212
- 60 A factory's production rate is $r(t) = 187 + 10t$ units/day, where t is days since a process change. Find the total production $P(15)$ given $P(0) = 0$.
- a. 2805
 - b. 3920
 - c. 3930
 - d. 3940
- 61 A factory's production rate is $r(t) = 108 + 30t$ units/day, where t is days since a process change. Find the total production $P(19)$ given $P(0) = 0$.
- a. 7437
 - b. 7467
 - c. 2052
 - d. 7497

- 62 A factory's production rate is $r(t) = 86 + 18t$ units/day, where t is days since a process change. Find the total production $P(14)$ given $P(0) = 0$.
- a. 2968
 - b. 1204
 - c. 2950
 - d. 2986
- 63 A factory's production rate is $r(t) = 184 + 25t$ units/day, where t is days since a process change. Find the total production $P(8)$ given $P(0) = 0$.
- a. 1472
 - b. 2272
 - c. 2247
 - d. 2297